

Vision Topics:

OpenCV: general purpose computer vision library. Does almost everything

- Free, open-source library (python and c++ apis)
- Includes
 - Feature detection
 - 3d reconstruction
 - Motion analysis
 - Optical flow
 - Photo manipulation / de-blur /
- What parts of openCV have already been used
 - Feature detection (SIFT, ORB)
 - Featurig matching
 - Takes 2 different images and matches key points

1. How are drones used for aerial imagery
 - a. Search and rescue, agriculture, law enforcement, environmentalism
 - b. Competitions
 - c. Types of aerial imagery
 - i. Oblique vs. Nadir
 1. Oblique is lower, while nadir is straight from above
 - ii. RGB-D / multispectral / plain RGB

Gimbal cameras will be used on our drones

OpenDroneMap(ODM)

- A lot simpler to use
- Has really nice APIS and even a web application

But also

- Not optimized for being done in flight

Next steps as a club

1. Make custom image stitching/SFM algo with OpenCV that can run in real time on the drone
 2. Dive deep into OpenDroneMap specifics and figure out how to neatly divide ip the stitching into small subcomponents, also real-time automatically
- Potential projects
 - SLAM (Simultaneous Localization and Mapping)
 - optical flow
 - Structure from Motion (getting a terrain map from image data)
 - Applying drone imagery

Skill Booster (Description on slides)

1. Color me impressed
2. I'll be Needin Stitches